# Work Summary

The project developed and tested a low-cost way to detect unusual smart-home device behaviour and limit what a compromised device can reach. A device can still appear to work normally while probing computers or storage devices on the same network, making suspicious activity hard for home and small-office users to spot.

We documented the network and captured about 24 hours of traffic in each setup, before and after segmentation. We used Zeek and Python to build traffic baselines for five devices from their destinations, protocols and traffic volumes. The final VLAN and firewall configuration blocked the unauthorised IoT-to-LAN paths we tested. We also traced dropped Matter status reports to a connection-tracking timeout and adjusted it to reduce those drops.

We completed the lateral scanning, DNS bypass, bulk upload and slow upload tests. The detection rules flagged all four test cases. We also completed the Splunk dashboard and ran a separate Mininet lab to compare network exposure under different configurations. All 28 documented device-function checks passed, including local control, remote control and recovery after a router reboot. One smart plug needed a power cycle after a wireless restart.

The portfolio includes sanitised packet captures, analysis scripts, device profiles, configuration records, test results and a dashboard guide. Students and small-network administrators can use these materials to trace alerts back to the underlying evidence and repeat similar assessments. Further validation could measure false alerts on a separate sample of normal traffic and have a second practitioner follow the instructions independently.